Record: It consists of a group of data items related to an object of data processing e.g., a payroll, record of an employee which contains data fields as name, age, qualification, sex, wage rate etc.

Data file is a compilation of related data records maintained in some pre-arranged order. Files are usually created function-wise e.g., a payroll file might consist of 1,000 individual employees data records and the records might be arranged in file by employee number.

The relationship of a character, data item, record and the file is shown in figure 1 below

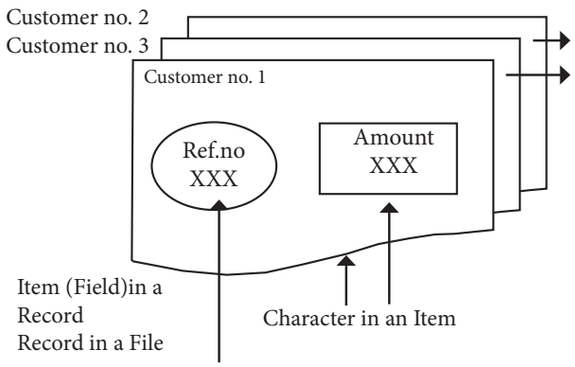

Fig.6.1. Relationship of Character, Data, Record and File

6.2. Types of Data:

Data are broadly divided into two types namely quantitative (or numeric) and qualitative:

1) **Quantitative Data:** Quantitative or numerical data refers to the observations that can be counted or measured. The data are said to be discrete if the measurements are integers (e.g. number of members in a family, number of students in the classroom, number of chocolates consumed per day) and continuous if the measurements can take on any value, usually within some range (e.g. weight).

2) **Qualitative Data:** Qualitative data is descriptive information (describes something). It arises when the observations fall into separate distinct categories. Since qualitative data always have a limited number of alternative values, such variables are also described as discrete.

Examples are: Colour of eyes: blue, green, brown etc

Exam result: pass or fail

Socio-economic status: low, middle or high.

Qualitative Data is broken down into Two Sub-Types:

- nominal if there is no natural order between the categories (e.g. eye colour), or
- Ordinal if an ordering exists (e.g. exam results, socio-economic status).

6.3. Data Processing – Meaning:

Data processing means taking raw data (facts and figures) and processing them manually or with the help of machines to produce, organized and useful information. It is restructuring \

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